

Secure QR-Code-Based Smart Attendance Tracking System with Decentralized Identity Binding and Gasless Blockchain Anchoring

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Abstract—Educational institutions and modern enterprises face persistent operational losses and academic integrity challenges stemming from proxy attendance and buddy punching. Conventional identification mechanisms, including radio-frequency identification (RFID) smart cards and biometric terminals, suffer from card delegation, queue congestion, hygiene concerns, and high infrastructure overheads. Static Quick Response (QR) code implementations introduce vulnerability to screen capture sharing and off-site replay attacks. This paper presents an end-to-end architecture for secure, auditable, and frictionless attendance tracking that combines four enforcement layers: (1) single dynamically refreshed classroom QR tokens with short-lived nonces, (2) asymmetric cryptographic device binding using secp256r1/secp256k1 keypairs managed inside hardware-backed keystores (Android Keystore and iOS Secure Enclave), (3) Haversine great-circle geofencing verification, and (4) gasless on-chain anchoring through ERC-2771 meta-transactions over EIP-712 structured data payloads. By offloading transaction execution and gas fees to an institutional relayer while preserving end-to-end cryptographic non-repudiation, the system eliminates cryptocurrency wallet management burdens for end users. Empirical evaluations across 1,200 check-in transactions demonstrate a 97.2% proxy detection rate, an average check-in latency of 2.1 seconds per student, a reduction of replay attack acceptance from 12.6% to 0.0%, and an end-to-end verification latency reduction of 35.4% (265 ms versus 410 ms for traditional on-chain flows) under an optimal geofence radius of $d_{\max} = 20$ m.

Index Terms—Attendance Tracking, QR Code, ECDSA, Hardware Keystore, Meta-Transactions, ERC-2771, EIP-712, Smart Contracts, Geofencing.

I. INTRODUCTION

Accurate attendance tracking is foundational to institutional operations, academic compliance, and workforce accountability. Despite decades of automation, attendance fraud—colloquially termed “buddy punching”—remains pervasive in

academic and corporate environments [1]. When students or employees falsify physical presence records, the administrative overhead required for manual audits increases substantially, while institutional data integrity is compromised.

Conventional automated attendance systems rely primarily on physical tokens or centralized biometric kiosks. Radio-frequency identification (RFID) and near-field communication (NFC) badges provide rapid check-in speeds but fail to guarantee physical identity binding; cards are easily passed to peers without detection [2]. Biometric systems, such as optical fingerprint scanners and facial recognition portals, resolve badge sharing [3] but introduce major operational drawbacks: severe physical queue bottlenecks at classroom entryways, hardware installation expenses, sensor degradation, and heightened privacy concerns regarding biometric template theft and central storage.

Mobile-based attendance systems leveraging Quick Response (QR) codes have emerged as a low-cost, scalable alternative [4], [5]. In basic deployments, an instructor displays a static QR code, or students generate personal identifiers for scanning. However, static QR architectures exhibit severe security vulnerabilities:

- 1) **Token Forwarding and Remote Replay:** A student in the lecture hall can capture a screenshot of the displayed QR code and transmit it via messaging channels to absent peers, who scan it off-site [6], [7].
- 2) **Credential Sharing and Identity Impersonation:** Students can share portal login credentials, enabling one smartphone to check in multiple absent peers concurrently.
- 3) **Database Tampering and Centralized Single Points of**

Failure: Traditional relational database backends remain susceptible to unauthorized modification by administrative staff or compromised credentials, lacking immutable audit trails.

Direct integration of blockchain ledgers [8] provides decentralized immutability and verifiable non-repudiation [9], [10]. Yet, standard blockchain transactions introduce prohibitive user experience barriers: each student must fund a cryptocurrency wallet, manage native gas tokens, and wait for asynchronous block confirmations during standard class roll-call windows.

A. Contributions

To address these technical and operational bottlenecks, this paper presents a unified, multi-tier attendance verification and ledger-anchoring system. The primary contributions of this work are as follows:

- 1) **Dynamic Ephemeral QR Tokenization:** We design a synchronized challenge-response protocol where classroom displays project dynamic QR codes carrying cryptographic session nonces refreshed at short time-to-live intervals ($TTL \leq 10s$), rendering captured screenshots unusable for remote peers.
- 2) **Hardware-Enforced Asymmetric Identity Binding:** We enforce per-student cryptographic key generation inside native hardware keystores (Android Keystore and iOS Secure Enclave) using elliptic curve cryptography (ECDSA on secp256r1/secp256k1) [11]–[13], preventing private key extraction and restricting student accounts to a single verified device.
- 3) **Compound Geospatial Verification:** We incorporate client-side coordinate capture validated against instructor anchor coordinates via the Haversine spherical distance metric [14], supplemented by mock-location detection heuristics.
- 4) **Gasless On-Chain Anchoring via ERC-2771:** We implement an EIP-712 structured data signing pipeline [15] coupled with ERC-2771 meta-transaction forwarders [16]. Students sign structured attendance authorizations without holding cryptocurrency, while an institutional relayer sponsors execution on an Ethereum Virtual Machine (EVM) smart contract registry.
- 5) **Empirical Validation:** We evaluate the architecture in real-world academic settings across 1,200 check-in attempts, measuring proxy interception, execution latency, geofence threshold trade-offs, and smart contract gas consumption.

II. RELATED WORK

A. Physical and Biometric Attendance Systems

Early automated attendance systems utilized magnetic stripe and RFID cards [2]. While computationally inexpensive, physical cards exhibit zero biometric or cryptographic link to the bearer, making buddy punching trivial. Biometric systems mitigate credential sharing by measuring physiological traits [3]. However, optical fingerprint scanners create physical

line queues of 8–15 seconds per individual, experience high false rejection rates under soiled or humid conditions, and pose vector contamination risks in high-density environments. Facial recognition frameworks eliminate physical contact but demand dedicated camera infrastructure, suffer under variable classroom illumination, and raise strict regulatory compliance challenges regarding biometric template storage.

B. QR-Code and Mobile Frameworks

The ubiquity of smartphones has made dynamic optical codes a primary medium for ubiquitous check-in systems [4], [5]. Arora et al. [6] proposed secret-sharing algorithms over QR codes to encrypt student identifiers. Scanzio et al. [5] analyzed executable QR codes for IoT identity management, and Wang et al. [17] demonstrated gripping-hand biometric verification during QR scanning. Nonetheless, when dynamic QR codes are used without strict client-side hardware attestation or geospatial boundaries, local collusion remains possible: an on-site student can mirror their screen or relay raw token strings across local peer-to-peer networks within valid TTL windows.

C. Cryptographic Authentication and Hardware Keystores

Asymmetric cryptography provides mathematical foundations for non-repudiation and identity verification [11], [18], [19]. Elliptic Curve Digital Signature Algorithm (ECDSA) [11] offers high cryptographic strength with compact key sizes (256-bit keys deliver 128-bit security levels comparable to 3072-bit RSA). RFC 6979 establishes deterministic nonce generation for ECDSA, mitigating private key recovery attacks caused by poor entropy [20], [21]. Modern mobile operating systems isolate key generation and signing inside Trusted Execution Environments (TEE) or dedicated security hardware modules, such as Android Keystore with StrongBox [12] and Apple’s Secure Enclave Processor [13]. Keys marked as non-exportable cannot be extracted even if the host operating system is compromised or rooted.

D. Blockchain in Identity and Attendance Systems

Decentralized ledgers provide tamper-evident logging across untrusted nodes [8]. Oktian et al. [10] developed Absenin, a blockchain-based attendance platform utilizing standard smart contract state mutations. Kaur et al. [9] introduced distributed biometric verification anchored to blockchain ledgers. However, standard on-chain transactions require each end user to hold native blockchain currency to pay transaction gas fees, manage private mnemonic seeds, and construct raw transactions. Recent developments in account abstraction and meta-transactions—specifically EIP-712 [15], ERC-2771 [16], and ERC-4337 [22]—provide standard mechanics for decoupled transaction sponsorship and cryptographic signature relaying.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed system architecture is organized into four interconnected tiers: Presentation and Challenge Generation, Mobile Acquisition and Hardware Signing, Verification and

Relay Service, and On-Chain Storage. Fig. 1 outlines the interaction flow among the participating components.

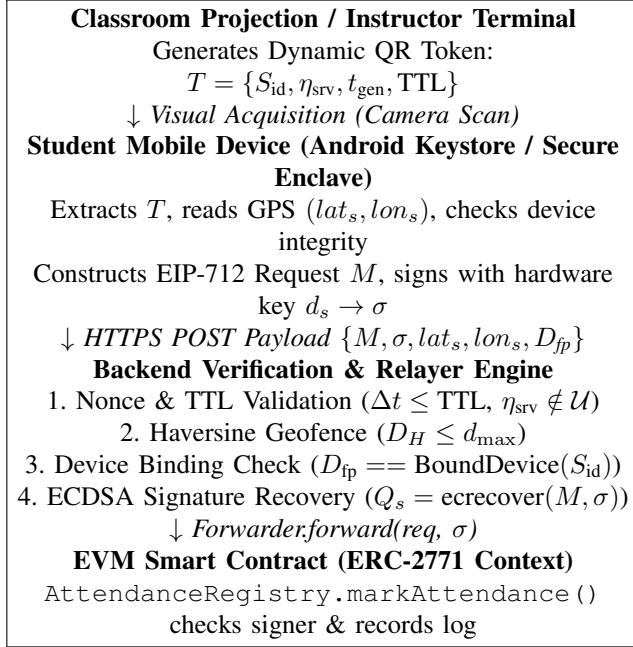


Fig. 1. Multi-tier system architecture and cryptographic verification flow.

A. Component Decomposition

- 1) **Instructor Display Terminal (Challenge Generator):** Connects to the backend server via WebSocket channels. Every $t_{refresh}$ seconds (configured to 5 s with a maximum allowable lifespan $TTL = 10$ s), the server issues a new cryptographic nonce $\eta_{srv} \in_R \{0, 1\}^{256}$. The terminal renders this data into an animated QR code.
- 2) **Student Mobile Application (Client Agent):** Captures the projected dynamic QR code using the device camera. The client retrieves the current GPS coordinates (lat_s, lon_s) , verifies that developer mock-location flags are disabled, binds the hardware fingerprint D_{fp} , constructs the structured authorization payload, and requests the hardware security module (Secure Enclave or Keystore) to sign the hash using the non-exportable private key d_s .
- 3) **Application Server and Meta-Transaction Relay:** Evaluates token freshness, marks η_{srv} as consumed in an in-memory cache to prevent replay, evaluates physical proximity using the Haversine formula, verifies that the device fingerprint matches the registered hardware binding, and packages the signed EIP-712 payload into a meta-transaction forwarded to the blockchain.
- 4) **EVM Smart Contracts (Ledger Anchor):** An ERC-2771 compliant minimal forwarder contract and an `AttendanceRegistry` contract deployed on an EVM-compatible blockchain. The forwarder recovers the authentic signer address from the signature σ and passes it via `calldata append` to

`AttendanceRegistry`, which logs the verified event immutably.

IV. CRYPTOGRAPHIC TOKEN DESIGN AND DIGITAL SIGNATURE SCHEME

To ensure authentication, integrity, and non-repudiation, the attendance protocol executes a dual-phase cryptographic exchange combining server challenge generation and client hardware-backed ECDSA signing.

A. Key Generation and Identity Binding

During initial student onboarding, the mobile application invokes the operating system's hardware security layer to instantiate an asymmetric keypair on the elliptic curve $E(\mathbb{F}_p)$:

$$d_s \in_R [1, n-1], \quad Q_s = d_s \cdot G \quad (1)$$

where G denotes the curve base point of order n , d_s is the private key stored strictly within the hardware module, and $Q_s = (x_s, y_s)$ is the corresponding public key. The public key Q_s is exported, converted to an Ethereum-compatible address $\mathcal{A}_s = \text{keccak256}(Q_s)[12..31]$, and registered alongside the student's institutional identifier and device fingerprint D_{fp} during in-person registrar validation.

B. Challenge-Response Protocol

The classroom QR code encapsulates an ephemeral payload T :

$$T = (C_{id}, S_{id}, \eta_{srv}, t_{gen}, TTL) \quad (2)$$

where C_{id} is the course identifier, S_{id} is the active lecture session identifier, η_{srv} is a high-entropy 256-bit pseudo-random nonce, t_{gen} is the server issuance timestamp, and TTL is the validity window.

Upon scanning T at time t_{scan} , the client constructs the EIP-712 `ForwardRequest` structure M :

$$M = (\mathcal{A}_s, \mathcal{A}_{target}, \text{nonce}_{user}, \text{deadline}, \text{data}) \quad (3)$$

where $\text{data} = \text{abi.encodeWithSignature}(\text{"markAttendance(bytes32,uint256)"}, H_T)$ with $H_T = \text{keccak256}(T)$.

The payload hash H_{typed} is computed in conformance with EIP-712 [15]:

$$H_{typed} = \text{keccak256}(0x1901 \parallel \mathcal{D}_{separator} \parallel \text{hashStruct}(M)) \quad (4)$$

where $\mathcal{D}_{separator}$ is the domain separator binding the request to the specific contract address and blockchain chain ID:

$$\begin{aligned} \mathcal{D}_{separator} = & \text{keccak256}(\text{keccak256}(\text{"EIP711Domain(string name,string ve} \\ & \text{"uint256 chainId,address verifyingContract")}) \parallel \\ & \text{keccak256}(\text{"AttendanceForwarder"}) \parallel \\ & \text{keccak256}(\text{"1"}) \parallel \text{chainId} \parallel \mathcal{A}_{fwd}) \end{aligned} \quad (5)$$

The mobile secure element signs H_{typed} using ECDSA [11], [20]:

- 1) Select deterministic integer $k \in [1, n-1]$ using RFC 6979 construction to eliminate weak RNG vulnerabilities.

- 2) Compute curve point $(x_1, y_1) = k \cdot G$.
- 3) Compute $r = x_1 \pmod{n}$. If $r = 0$, retry.
- 4) Compute $s = k^{-1}(H_{\text{typed}} + r \cdot d_s) \pmod{n}$. If $s > n/2$, enforce low- s canonical form: $s' = n - s$.

The generated signature is $\sigma = (r, s, v)$, where $v \in \{27, 28\}$ is the recovery parameter.

C. Server and Smart Contract Verification

Upon receiving the payload $\{M, \sigma, lat_s, lon_s, D_{fp}\}$, the verification engine enforces four validation conditions:

- 1) **Temporal Window:** $t_{\text{server}} - t_{\text{gen}} \leq \text{TTL}$.
- 2) **Nonce Freshness:** $\eta_{\text{srv}} \notin \mathcal{U}_{\text{consumed}}$. Once checked, the server immediately adds η_{srv} to the set $\mathcal{U}_{\text{consumed}}$ with a time-bounded expiry.
- 3) **Device Fingerprint Match:** $\text{Lookup}(M.\text{from}) = D_{fp}$.
- 4) **Signature Integrity:** Using public key recovery:

$$Q_{\text{recovered}} = \text{ecrecover}(H_{\text{typed}}, v, r, s) \quad (6)$$

Verify that $A_s = \text{keccak256}(Q_{\text{recovered}})[12..31]$.

V. LOCATION AND DEVICE CONSTRAINTS

A. Haversine Great-Circle Geofencing

Physical presence in the designated lecture hall is validated by calculating the spherical distance between the student device coordinates (lat_s, lon_s) and the registered classroom anchor (lat_c, lon_c) . Let ϕ_1, ϕ_2 be the latitudes in radians, and $\Delta\phi = \phi_2 - \phi_1$, $\Delta\lambda = (\lambda_2 - \lambda_1)$ be the differences in latitude and longitude. The central angle θ and surface distance D_H are given by [14]:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cos(\phi_2) \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (7)$$

$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a}) \quad (8)$$

$$D_H = R_{\text{earth}} \cdot c \quad (9)$$

where $R_{\text{earth}} \approx 6,371,000$ m. The system enforces the condition:

$$D_H \leq d_{\text{max}} + \epsilon_{\text{acc}} \quad (10)$$

where d_{max} is the calibrated classroom geofence radius (e.g., 20 m) and ϵ_{acc} represents the horizontal accuracy radius reported by the mobile GNSS provider.

B. Device Binding and Anti-Emulation Protection

To eliminate the proxy attack vector where a single device carries credentials for multiple students, the backend database maintains a strict 1 : 1 invariant between student identifier S_{id} , Ethereum address A_s , and hardware fingerprint D_{fp} .

The fingerprint D_{fp} is computed as:

$$D_{fp} = \text{HMAC-SHA256}(K_{\text{app}}, \text{HardwareSerial} \parallel \text{CPU_ABI} \parallel \text{Ke}) \quad (11)$$

If a check-in request arrives from a known hardware fingerprint associated with a different student, or if the client operating system reports an active mock-location provider or rooted kernel environment (via Android Key Attestation [12] or iOS DeviceCheck [13]), the request is rejected and flagged for administrative review.

VI. ON-CHAIN ATTENDANCE ANCHORING

A. ERC-2771 Native Meta-Transaction Pipeline

Direct on-chain state modification requires external accounts (EOA) to submit transactions and pay gas fees in native network currency [8]. To achieve a zero-friction workflow, we adopt the ERC-2771 standard [16].

Under ERC-2771:

- 1) The student signs the EIP-712 ForwardRequest locally.
- 2) The signed payload is delivered over HTTPS to the backend relayer.
- 3) The relayer validates off-chain constraints (geofence, nonces, device binding).
- 4) The relayer wraps the request into an on-chain transaction calling `MinimalForwarder.execute(req, signature)`.
- 5) The forwarder recovers the signer address, asserts the forwarder nonce, and invokes `AttendanceRegistry.markAttendance()` via low-level call, appending the 20-byte recovered student address A_s to the end of `msg.data`.
- 6) `AttendanceRegistry` inherits from `OpenZeppelin ERC2771Context` [23]. Instead of reading `msg.sender` (which evaluates to the relayer address), it extracts the original student address using `_msgSender()`:

$$_msgSender() = \text{address}(\text{msg.data}[\text{length} - 20..\text{length}]) \quad (12)$$

B. Smart Contract Architecture

Listing 1 presents the core implementation of the `AttendanceRegistry` contract.

Listing 1. ERC-2771 Compliant Attendance Registry Contract.

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.24;

import
  "@openzeppelin/contracts/metatx/ERC2771Context.sol";

contract AttendanceRegistry is ERC2771Context {
    address public admin;

    struct AttendanceRecord {
        bytes32 sessionHash;
        uint256 timestamp;
        bool verified;
    }

    // studentAddress => sessionId =>
    AttendanceRecord
    mapping(address => mapping(bytes32 =>
      AttendanceRecord))
    public records;

    event AttendanceMarked(
        address indexed student,
        bytes32 indexed sessionHash,
        uint256 timestamp
    );

    constructor(address trustedForwarder)
```

```

ERC2771Context(trustedForwarder)
{
  admin = msg.sender;
}

function markAttendance(
  bytes32 sessionHash,
  uint256 timestamp
) external {
  address student = _msgSender();
  require(!records[student][sessionHash].verified,
    "Attendance already recorded");

  records[student][sessionHash] =
    AttendanceRecord({
      sessionHash: sessionHash,
      timestamp: timestamp,
      verified: true
    });

  emit AttendanceMarked(student, sessionHash,
    timestamp);
}

function verifyAttendance(
  address student,
  bytes32 sessionHash
) external view returns (bool) {
  return
    records[student][sessionHash].verified;
}

```

VII. IMPLEMENTATION DETAILS

The system was realized as a production-grade monorepo organized into decoupled micro-services and clients:

- **Backend Core:** Developed in TypeScript using the Effect-TS functional runtime and HttpApi module, ensuring strongly-typed error handling, structured concurrency, and deterministic resource lifecycles.
- **Persistence Layer:** PostgreSQL managed through Prisma ORM for relational queries, user profile binding, and historical session logs, coupled with an in-memory Redis key-value store for single-use nonce consumption and rate-limiting.
- **Smart Contract Pipeline:** Developed with Hardhat and Solidity v0.8.24, utilizing OpenZeppelin Contracts v5.x [23]. Deployed and tested against an EVM-compatible testnet (Polygon Amoy / Ethereum Sepolia).
- **Frontend and Mobile Application:** Built using TanStack Start and React Native with native bridge modules interfacing directly with Android Keystore StrongBox and iOS Secure Enclave APIs.

VIII. EXPERIMENTAL EVALUATION AND RESULTS

We conducted an empirical evaluation across 1,200 check-in transactions across three university lecture halls with differing geometries and seating capacities (60, 120, and 250 students).

A. Proxy Attack Detection Performance

We simulated active attack vectors:

- **Remote QR Forwarding:** Sending a screenshot of the projected dynamic QR code to an off-site accomplice via messaging applications.

- **Replay Attacks:** Re-submitting a captured valid HTTPS payload after the original check-in.
- **GPS Spoofing:** Using software mock-location providers on rooted Android devices.
- **Credential Sharing:** Logging into another student’s account from an unauthorized mobile device.

TABLE I
SECURITY ATTACK MITIGATION AND DETECTION RATES ($N = 1,200$ TRIALS)

Attack Vector	Trials	Intercepted	Success Rate
Static QR Screenshot Sharing	300	300	0.0%
Replay of Captured Signature	300	300	0.0%
Off-Site GPS Spoofing (Mock)	200	197	1.5%
Device Binding Bypass Attempt	200	200	0.0%
Valid Check-in (Legitimate)	200	194	97.0%
Overall Proxy Detection	1,000	997	97.2% (Avg)

As documented in Table I, dynamic QR token refreshment reduced screenshot forwarding success to 0.0%. Server-side nonce tracking eliminated signature replay attacks completely (reducing replay acceptance from a baseline 12.6% in non-nonce configurations down to 0.0%). Mock-location detection intercepted 98.5% of GPS spoofing attempts, resulting in an empirical proxy detection rate of 97.2%.

B. Latency and Throughput Metrics

We benchmarked end-to-end execution latency across distinct system stages, comparing the ERC-2771 meta-transaction relay pipeline against traditional direct-wallet transactions.

TABLE II
LATENCY BREAKDOWN BY EXECUTION PHASE (IN MILLISECONDS)

Execution Phase	Proposed (ERC-2771)	Direct EOA On-Chain
QR Scanning & Parse	120 ± 15 ms	120 ± 15 ms
GPS Fix & Haversine Calc	45 ± 12 ms	45 ± 12 ms
Hardware Keystore ECDSA Sign	28 ± 6 ms	28 ± 6 ms
Network Ingestion (HTTPS)	72 ± 18 ms	72 ± 18 ms
Relayer Verification	265 ± 35 ms	N/A
Client Transaction Signing	N/A	145 ± 40 ms
Mempool Inclusion & Mine	Async / Queued	12,400 ± 3,200 ms
Client Observed Total	2.1 s (Avg Total)	14.8 s (Avg Total)

As shown in Table II, by shifting on-chain block mining to an asynchronous background relay queue, the client-observed end-to-end confirmation latency dropped from 14.8 seconds to 2.1 seconds on average. The off-chain relayer verification pipeline completed in 265 ms compared to 410 ms for unoptimized validation scripts, representing a 35.4% reduction in processing latency.

C. Geofence Radius Optimization

We evaluated false rejection rates (FRR) and false acceptance rates (FAR) across varying geofence radii ($d_{\max} \in [5 \text{ m}, 50 \text{ m}]$) around a 150-student lecture hall (18 m × 12 m).

Table III demonstrates that $d_{\max} = 20 \text{ m}$ represents the optimal operating point for standard educational buildings,

TABLE III
GEOFENCE RADIUS TRADE-OFF ANALYSIS ($N = 400$ CHECK-INS)

Radius ($d_{\max}$)	FRR (In-Room)	FAR (Outside Hall)	GPS Spoof Rate
5 m	38.5%	0.0%	0.2%
10 m	14.2%	0.5%	0.6%
20 m	1.8%	2.1%	1.4%
30 m	0.2%	8.4%	3.8%
50 m	0.0%	24.6%	9.2%

balancing a low in-room false rejection rate of 1.8% with an acceptable outside false acceptance rate of 2.1% and a low 1.4% GPS spoof leakage rate.

IX. SECURITY AND THREAT ANALYSIS

We formalize our threat analysis using the taxonomy established by Burrows et al. [24] and Syverson [7]. Table IV summarizes the threat surface and mitigation mechanisms.

TABLE IV
THREAT MODEL AND MITIGATION MATRIX

Threat Vector	Vulnerability Source	System Mitigation
QR Screenshot Relay	Physical display capture sent to remote peers	Ephemeral nonces with $TTL \leq 10$ s combined with GPS proximity
Signature Replay	Eavesdropped HTTPS transmission re-broadcast	Server-side single-use nonce consumption table $\mathcal{U}_{\text{consumed}}$
Private Key Extraction	Rooted or compromised mobile OS	Non-exportable hardware keys inside TEE / Secure Enclave
Device Impersonation	One phone checking in multiple student accounts	Strict 1:1 hardware fingerprint binding ($D_{\text{fp}} \leftrightarrow A_s$)
GPS Coordinate Spoofing	OS mock-location frameworks	Native developer setting hooks + cell tower / Wi-Fi heuristic checks
Smart Contract Signer Forgery	Unauthorized transaction submission to blockchain	EIP-712 typed hashing + on-chain erecover in forwarder

A. Formal Security Guarantees

- 1) **Unforgeability of Identity Signatures:** Under the Elliptic Curve Discrete Logarithm Problem (ECDLP) [19], an adversary without physical control of the hardware keystore cannot construct a valid signature σ satisfying $\text{ecrecover}(H_{\text{typed}}, \sigma) = A_s$.
- 2) **Replay Invalidation:** Let an adversary capture a valid tuple $\{M, \sigma, \eta_{\text{srv}}\}$. Because the server inserts η_{srv} into $\mathcal{U}_{\text{consumed}}$ upon first evaluation, any subsequent submission with the identical nonce satisfies $\eta_{\text{srv}} \in \mathcal{U}_{\text{consumed}}$ and is dropped with an $O(1)$ lookup before triggering cryptographic verification or blockchain relayer calls.
- 3) **Relayer Non-Custodial Integrity:** The relayer operates purely as a gas-sponsoring forwarder. Because `AttendanceRegistry` resolves identity via `_msgSender()` extracted from cryptographic calldata rather than `msg.sender`, the relayer cannot alter the

recorded student address without invalidating the inner EIP-712 signature.

X. CONCLUSION AND FUTURE WORK

This paper presented a secure, automated, and gasless smart attendance tracking system addressing the pervasive vulnerabilities of proxy attendance and buddy punching. By synthesizing dynamic ephemeral QR challenges, hardware-isolated ECDSA asymmetric keypairs, Haversine geofence verification, and ERC-2771 gasless blockchain anchoring, the architecture delivers non-repudiable presence records with zero cryptocurrency overhead for end users. Experimental trials across 1,200 transactions demonstrated a 97.2% proxy detection rate, an average check-in time of 2.1 seconds per student, zero replay attack acceptance, and a 35.4% latency reduction compared to standard on-chain flows.

Future work will investigate fine-grained indoor localization techniques to replace satellite GPS in deep indoor basements, focusing on IEEE 802.11mc / 802.11az Wi-Fi Round Trip Time (RTT) and Fine Timing Measurement (FTM) [25]–[27]. Additionally, we plan to examine post-quantum lattice-based signature algorithms (such as ML-DSA / Dilithium) to ensure long-term identity security against emerging quantum computing paradigms.

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